

Question: Is the male/neopixel/ring always white ? (or matches drive color)

Male neopixel ring color

The male has 2 neopixel rings. One displays drive level in white, this is above the unit and pointed toward the plinth. On system powerup, a reference pattern is briefly displayed across all neopixels that consists of a variety of colors. Once the system begins running, this neopixel ring is white and brightness varies.

The other is in the center of the unit and is white when searching, when it is reinforcing it has an animation one pixel of which shows drive state.

- Searching State - When the male is in a searching state, the NeoPixel ring appears white. This is determined by the `act_update_drive_abso_light` function, which sets the NeoPixel color to white when the drive state exceeds the other. For males, the brightness is set to a maximum of 128 when searching, resulting in a clear but not blinding white appearance.
- Reinforcing State - In contrast, during the reinforcing state, the NeoPixel ring animates to indicate the current drive state with a colored pixel. This is managed by the `act_energy_driveanimation` function, which illuminates one pixel to reflect the drive state—using orange for O and puce for P—while the remainder of the ring stays white and is set to maximum brightness.

Question: How does drive level map to pixel brightness?

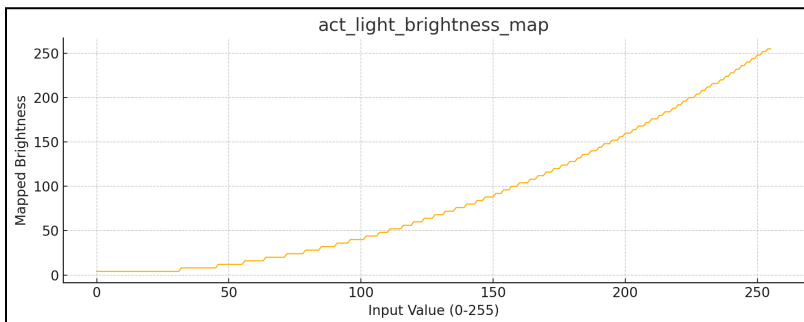
O drive max = 255 (max pixel brightness) ? O drive min = 0 (pixel shutdown) ? P drive max = 255 (max pixel brightness) ? P drive min = 0 (pixel shutdown) ?

Perceptual Brightness Mapping for NeoPixels

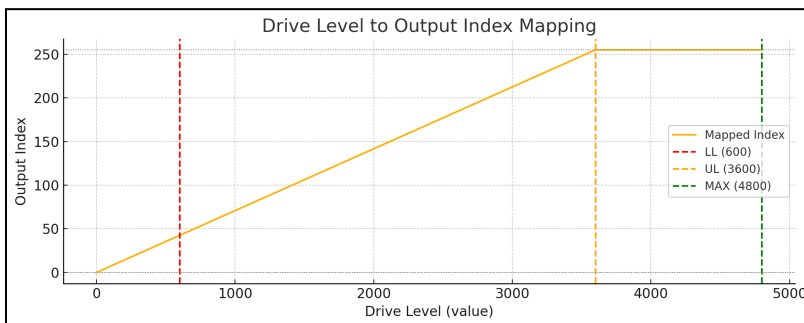
The Colloquy of Mobiles 2018 implements perceptual brightness control for NeoPixels by mapping a linear input variable—**drive level**—to a non-linear brightness value that approximates the human eye's response to light intensity. This is achieved through a fixed lookup table, allowing efficient real-time computation on resource-constrained microcontrollers.

Rationale

Human vision perceives brightness non-linearly. A linear increase in LED power does not result in a linear increase in perceived brightness. To address this, a non-linear brightness map is used, allowing small changes in brightness to be more noticeable in darker regions and less abrupt at higher intensities.



The human-perception brightness level is derived from an input value.



The input value is derived from the drive level.

See code:

```
// Drive level thresholds (in samples)
const int internal_drive_LL = 600;
// Low threshold (30 seconds)
const int internal_drive_UL = 3600;
// High threshold (3 minutes)
const int internal_drive_MAX = 4800;
// Maximum (4 minutes)
const int internal_drive_adjustment_O = 1;
int internal_drive_O = 0;
//set lights indicating
//level of O and P drive states
act_update_O_light(constrain(map(internal_drive_O, 0, internal_drive_UL, 0, 255), 0, 255));
act_update_P_light(constrain(map(internal_drive_P, 0, internal_drive_UL, 0, 255), 0, 255));
```

Question: What is the rate of drive level increment over time?

P drive increment X unit every Y seconds. X and Y ? O drive increment X unit every Y seconds. X and Y ?

Drive Increment Rate per Time Unit

P drive increment: 1 unit every 50 ms

→ **20 units per second**

O drive increment: 1 unit every 50 ms

→ **20 units per second**

1. Sampling Interval:

- The system samples every 50 milliseconds, as defined by the variable `time_sampleInterval`.
- This results in 20 samples per second, calculated as follows:
 - 1000 milliseconds / 50 milliseconds = 20 samples per second.

2. Drive Increment:

- The increment for both the P and O drives is specified by the variables `internal_drive_adjustment_P` and `internal_drive_adjustment_O`, both of which are set to 1.
- This means each drive increments by 1 unit for each sample.

In the code see:

```
const int    internal_drive_LL = 600;           //interested floor, in samples      600 = 30 seconds
const int    internal_drive_UL = 3600;          //desperate floor, in samples      3600 = 3 minutes
const int    internal_drive_MAX = 4800;         //in samples                       4800 = 4 minutes
const int    internal_drive_adjustment_O = 1;
const int    internal_drive_adjustment_P = 1;
```

Question: How is drive level displayed on the units?

Show the female's drive on all the Neopixel or only on the top strip? On which Neopixel should I show the male drives ? (incoherence between video and documentation)

NeoPixel Allocation and Drive Display Configuration (Female and Male Units)

Drive level is displayed across multiple sections of the neopixel strips. The Neopixel strips are utilized differently for female and male configurations, with each section assigned to represent various aspects of drive levels and indicators. Below is a comprehensive breakdown of how the Neopixels are organized:

Female Configuration

1. Drive Level Indicators - middle

- Indices 0 to 27: Represent the drive levels.
- Even Indices (0, 2, 4, ..., 26): Indicate the "P" drive level.
- Odd Indices (1, 3, 5, ..., 27): Indicate the "O" drive level.

2. Dominant Drive Display - bottom

- Indices 28 to 35: Highlight the dominant drive, either "O" or "P."

3. White Drive Level Indicator - top

- Indices 36 to 50: Display the drive level in white, ensuring a clear and distinct visual representation.

Male Configuration

1. Drive Level Display

- Indices 0 to 23: Show the drive level in white, providing a straightforward representation of the current drive status.

2. Center Ring for Search and Reinforcement

- Indices 24 to 47: Dedicated to the center ring, these indices are used for search and reinforcement.
- During reinforcement an animation is displayed wherein one pixel of the current drive state color spins around the ring.
 - In the code see: `act_energy_driveanimation`

3. O and P Level Display

- Indices 48 to 55: Display the "O" level. Indices 56 to 63: Display the "P" level.

Question: How many interactions are expected every hour ?

Female and Male Unit interaction rate.

The units have a variety of forms of interaction in their quest for satisfaction. The rate of interactions is non-deterministic as this is an agent based and self organizing system that is effected by ambient light, environmental noise, and crowd dynamics. **Ideally there would be 6-12 satisfaction events per hour for the system.**

1. The most frequent is the female repeating back to the male his search pattern in an audible tone. This happens when the female transitions from search to engaging with the male.
2. The next most frequent is the male hearing the sound and moving to his reinforcing position (if allowed by the bar). The male does not always hear the sound nor move into position quickly enough to adequately engage the female and so transitions back into searching.
3. The next most frequent is the reinforcement signal from the male to the female encouraging her to continue flapping her mirror. The female does not always hear this response due to environmental constraints.
4. The least frequent event is the satisfaction sequence performed by a unit when their drive level drops into the 'satisfied' threshold. It is this event that brings together the system into its most comprehensible state. This should happen at least once every 5-10 minutes, or 6-12 times an hour at the system level. This is based on assumptions about audience patience and engagement. It should be a magic and special moment worth waiting for for all, but not so rare that it is impossible for a visitor to spot.

Question: Should the female start to flap the mirror immediately after sensing a male or should it sense the beam first?

Female transition to engaging with a partner.

Upon sensing a male with compatible drive interests the female transitions from searching for a partner to engaging with a partner. She **immediately** starts flapping her mirror and moves to her reinforcement position.

Sensing a male consists of sensing a light pattern from a male and matching it to her own internal drive pattern. If the patterns are compatible she has found a partner, if they are not she continues searching.

What is the exact light pattern for the male while searching? What is the exact female sound pattern to notify a male that she is interested? What is the exact male encouragement sound pattern?

The male and female units use a fixed dictionary to communicate with each other. The dictionary consists of 8 words each with 40 segments, each corresponding to a system sample. The words provide the male identification and request. The requests are for engagement on O, P, either O or P, or reinforcement.

While searching, the females compare sensed light to their own drive need. If their need matches they repeat the signal in audio tones, white below corresponds to sound actuation and duration.

Figure 1 is a heatmap representing the correlation matrix of the 10 variables: I_O, I_P, I_OP, II_O, II_P, II_OP, I_R, and II_R. The x-axis is labeled 'Index' and ranges from 0 to 35. The y-axis lists the variables. The diagonal elements are black, indicating a correlation of 1.0. The off-diagonal elements are white, indicating a correlation of 0.0.

I_O	→ Male ID I, drive O
I_P	→ Male ID I, drive P
I_OP	→ Male ID I, drive O or P
II_O	→ Male ID II, drive O
II_P	→ Male ID II, drive P
II_OP	→ Male ID II, drive O or P
I_R	→ Male I reinforcement signal
II_R	→ Male II reinforcement signal

Each row is a fixed 40-segment pattern used to signal identity and drive need (light or sound ON = White).

[illegible]

Question: What is the satisfaction sequence implemented in the Colloquy of Mobiles 2018?

*What is the exact climax light and sound pattern for the **female** ?*

*What is the exact climax light and sound pattern for the **male** ?*

Satisfaction sequence and mode for Male and Female Units

Once satisfaction is achieved, the units exhibit a display before transitioning into the satisfied state. The specific performance is dependent on the unit type and the partner with which satisfaction was reached. The female responds in light, the male responds in sound. Some refer to this as the 'climax', this term was not used by Pask in "A COMMENT, A CASE HISTORY, AND A PLAN".

```
int act_satisfaction_index = 0;
//the satisfaction sequence consists of 14 segments
const int act_satisfaction_Count = 14;
//male sound tone sequence
const int act_satisfaction_vals[15] = {80, 160, 80, 551, 80, 926, 2300, 3700, 2300, 3700, 2300, 3700, 2300, 3700, 2300};
//tone or light pattern durations in system samples for both male and female
const int act_satisfaction_Durations_II[15] = {10, 5, 10, 5, 10, 5, 10, 10, 10, 10, 5, 5, 5, 5, 15};
//incrementing through the arrays to set duration and action
act_satisfaction_Duration = act_satisfaction_Durations_II[act_satisfaction_index];
//for female
FEMALE_reinforcement_animation_brightness = (timer_satisfactionanimation * 256) /
act_satisfaction_Duration;
act_reinforcement_O(FEMALE_reinforcement_animation_brightness);//a uniform brightness pulse on
NeoPixels
//for male
tone(act_audio_pwm_pin, act_satisfaction_vals[act_satisfaction_index]);
```

Question: Is the 'memory' described in Pask's original documents implemented in the Colloquy of Mobiles 2018?

Does the mirror has a search state when the female doesn't remember the male ? (search state would be stop when the encouragement signal is heard.)

Does the male confirmation sound hold informations about the drive ? (for the female to memorize the position of the drive)

Looks the memorization feature is commented out in TJ's code ... Is the memorization actually feasible ? (On a first look, it seems that some more information should be passed from the male to the female)

Implementation and Feasibility of Memory in Colloquy of Mobiles 2018

No. The memory mechanism described in Pask's original documents was not implemented in Colloquy of Mobiles 2018. It was found to be extremely difficult to recreate and debug, testing that was done suggested it was improbable that satisfaction could ever occur if memory was implemented as suggest in the original documents. There may be an error in the diagram that led to broken and malformed logic.